

Introduction, risk and financial institutions

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Risk management and financial institutions

Lecture 1.

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Course description

- weekly lectures and seminars
- individual assignment due by 18th May (10 points)
- group assignment due by 9th May (10 points)
- Final exam (80 points)
- 50% has to be achieved on all parts

Risk vs uncertainty

- *Risk*: more, but known possible future outcomes with given probabilities; the distribution is known.
- *Uncertainty*: more possible future outcome, but the probabilities or even all the possible outcomes are unknown.

Risk types

- Market risk
- Credit risk / partner risk
- Operational risk
- Liquidity risk
- Model risk
- Systemic risk
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Risk-return trade-off in investments

There is a trade off between risk and expected return

The higher the risk, the higher the expected return

Risk vs Return for Companies

If shareholders care only about systematic risk, should the same be true of company managers?

In practice companies are concerned about total risk

Earnings stability and company survival are important managerial objectives

The regulators of financial institutions are primarily interested in total risk

“Bankruptcy costs” arguments show that that managers may be acting in the best interests of shareholders when they consider total risk

History of bank regulation

1974: Basel Committee on Banking Supervision

1988: BIS Accord (Basel I.)

1996: Amendment to Basel I.

2004: Basel II.

- CRD I. 2006
- CRD II. 2009

2009: Basel II.5

- CRD III. 2010

2011: Basel III.

- CRD IV: Directive 2013/36/EU
- CRR: Regulation (EU) No 575/2013

Basel standards

Basel I. – credit risk

1996: Amendment – market risk of the trading book

2004: Basel II. – operational risk

2009: Basel II.5 – market risk revisited

Basel III. – liquidity risk, systemic risk

Measuring risk

A risk measure characterizes the distribution of the risk factor (return of the investment).

- Standard deviation /variance
- Quantile (percentile)
- Maximum /minimum
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Market risk – risk measurement

You are supposed to know.....

What does a cumulative distribution function show?

How do you define quantile?

Is it true that the correlation coefficients of two independent random variables equals zero?

And what about the reverse statement?

What are the main characteristics of normal distribution?

What is Geometric Brownian Motion?

The Question Being Asked in VaR

“What loss level is such that we are $X\%$ confident it will not be exceeded in N business days?”

VaR and Regulatory Capital

Regulators have traditionally used VaR to calculate the capital they require banks to keep, but have recently switched to ES for market risks

The market-risk capital has been based on a 10-day VaR estimated where the confidence level is 99%

Credit risk and operational risk capital are based on a one-year 99.9% VaR

Advantages of VaR

It captures an important aspect of risk
in a single number

It is easy to understand

It asks the simple question: “How bad can things
get?”

VaR vs. Expected Shortfall

VaR is the loss level that will not be exceeded with a specified probability

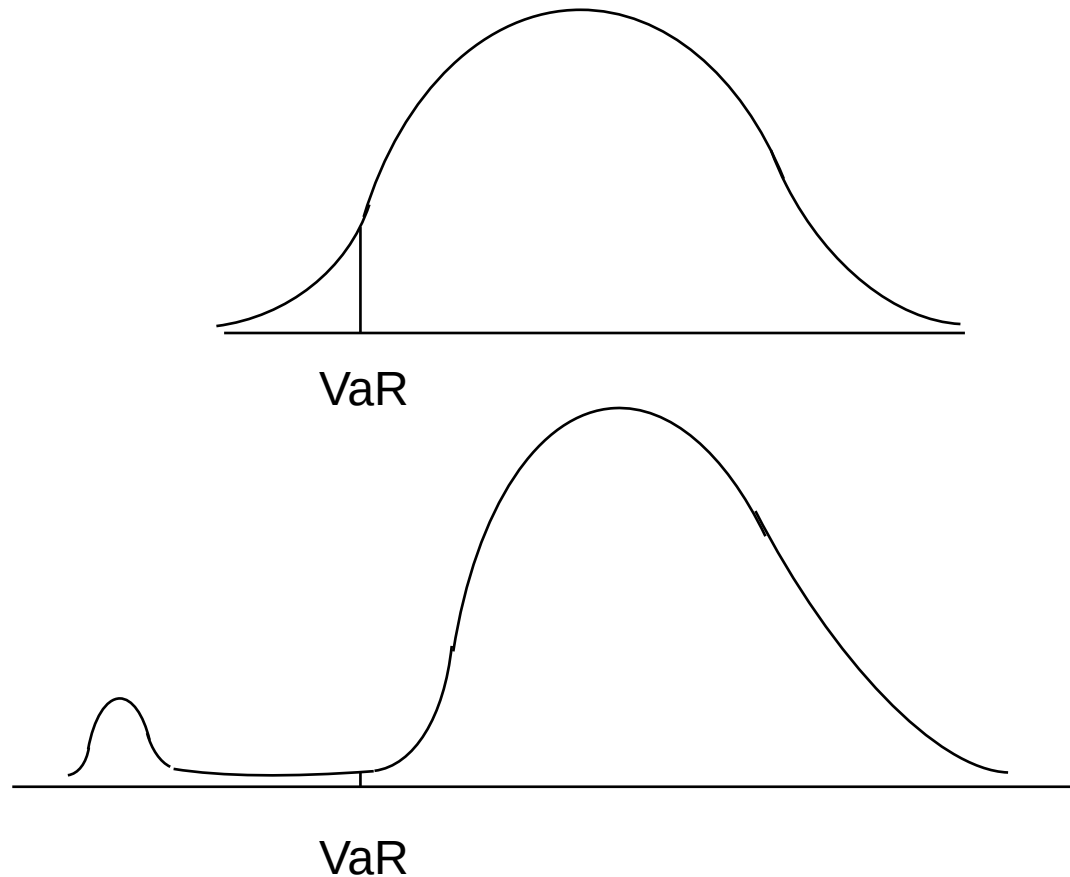
Expected shortfall (ES) is the expected loss of a given worst percentage of all outcomes.

If the distribution is continuous, ES is given that the loss is greater than the VaR level (also called C-VaR and Tail Loss)

Regulators have moved from using VaR to using ES for determining market risk capital

Two portfolios with the same VaR can have very different expected shortfalls

Distributions with the Same VaR but Different Expected Shortfalls



Coherent Risk Measures

Define a coherent risk measure as the amount of cash that has to be added to a portfolio to make its risk acceptable

Properties of coherent risk measure

- If one portfolio always produces a worse outcome than another its risk measure should be greater (monotonicity)
- If we add an amount of cash K to a portfolio its risk measure should go down by K (translation invariance)
- Changing the size of a portfolio by λ should result in the risk measure being multiplied by λ (positive homogeneity)
- The risk measures for two portfolios after they have been merged should be no greater than the sum of their risk measures before they were merged (sub-additivity)

Spectral Risk Measures

1. A spectral risk measure assigns weights to quantiles of the loss distribution
2. VaR assigns all weight to X th percentile of the loss distribution
3. Expected shortfall assigns equal weight to all percentiles greater than the X th percentile
4. For a coherent risk measure weights must be a non-decreasing function of the percentiles

Choice of VaR Parameters

Time horizon should depend on how quickly portfolio can be unwound. Regulators are planning to move toward a system where ES is used and the time horizon depends on liquidity. (See *Fundamental Review of the Trading Book*)

Confidence level depends on objectives. Regulators have used 99% for market risk and 99.9% for credit/operational risk.

A bank wanting to maintain a AA credit rating might use confidence levels as high as 99.97% for internal calculations.

Back-testing

Back-testing a VaR calculation methodology involves looking at how often exceptions (loss > VaR) occur

Alternatives: a) compare VaR with actual change in portfolio value and b) compare VaR with change in portfolio value assuming no change in portfolio composition

Suppose that the theoretical probability of an exception is p ($=1-X$). The probability of m or more exceptions in n days is

$$\sum_{k=m}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

Historical Simulation

Collect data on the daily movements in all market variables (known as risk factors)

The first simulation trial assumes that the percentage changes in all risk factors are as on the first day

The second simulation trial assumes that the percentage changes in all risk factors are as on the second day

and so on

Weighting of Observations

Let weights assigned to observations decline exponentially at rate λ as we go back in time

Rank observations from worst to best

Starting at worst observation sum weights until the required percentile is reached

The Model-Building Approach

The main alternative to historical simulation for calculating VaR or ES is to make assumptions about the probability distributions of the returns on the market variables

This is known as the model building approach (or sometimes the variance-covariance approach)

Normal Distribution Assumption

When losses (gains) are normally distributed with mean μ and standard deviation σ

$$\text{VaR} = \mu + \sigma Y$$

$$\text{ES} = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1-X)}$$

where X is the confidence level and

$$Y = N^{-1}(X)$$

Changing the Time Horizon

If losses in successive days are independent, normally distributed, and have a mean of zero

$$T \text{ - day VaR} = 1 \text{ - day VaR} \times \sqrt{T}$$

$$T \text{ - day ES} = 1 \text{ - day ES} \times \sqrt{T}$$

Extension

If there is autocorrelation ρ between the losses (gains) on successive days, we replace $\sqrt{T}$ by

$$\sqrt{T + 2(T - 1)\rho + 2(T - 2)\rho^2 + 2(T - 3)\rho^3 + \dots + 2\rho^{T-1}}$$

in these equations

Ratio of T -day VaR to 1-day VaR (Table 11.1)

	$T=1$	$T=2$	$T=5$	$T=10$	$T=50$	$T=250$
$\rho=0$	1.0	1.41	2.24	3.16	7.07	15.81
$\rho=0.05$	1.0	1.45	2.33	3.31	7.43	16.62
$\rho=0.1$	1.0	1.48	2.42	3.46	7.80	17.47
$\rho=0.2$	1.0	1.55	2.62	3.79	8.62	19.35

The Linear Model

We need to assume

The daily change in the value of a portfolio is linearly related to the daily returns from market variables

The returns from the market variables are normally distributed

The market variable are referred to as risk factors

Use of Actual Changes

For some variables such as interest rates and credit spreads it is usual to consider actual changes rather than proportional changes. The same formulas apply with the following definitions

Δx_i is the actual change in risk factor i

σ_i is the standard deviation of the change in the i th risk factor

δ_i is delta of portfolio with respect to the actual change in risk factor i

Thank you!
